

ClaimGuard project requirements and interface screenshots

BUAN 3301 · AI in Business · Meridian Insurance (fictional)

Purpose and scope

ClaimGuard is a single-page, end-user claim-review application, not an administrator dashboard. A reviewer selects one of 18 instructor-provided claims whose two reference photos were mapped by the team, or enters a new claim with photos. The app checks photo reuse across claims and searches for public-web matches with Google Cloud Vision Web Detection. A deterministic guardrail returns **Auto-approve**, **Escalate**, or **Blocked**. A match is a reason for review, not proof of fraud. Meridian Insurance is fictional and the app is a university-course prototype.

The application prepares a Google Form link with claim ID, claimant, photo-lookup summary, and decision. The app does not provide an email value or submit the Form. The Form currently collects **Verified** email addresses, requiring the respondent to sign in to Google; the person then clicks Submit.

End-to-end user flow

- 1. Choose a mapped reference claim, or enter a new claim's factual fields and description.
- 2. Review the reference claim's two mapped thumbnails, or add two to six new images by picker, drag-and-drop, or ZIP.
- 3. Run a separate lookup for each photo. The server checks cross-claim reuse and Web Detection, reporting successful matches, a successful zero-match result, or an unavailable web check distinctly.
- 4. The server reloads stored evidence, applies the guardrail, records an activity timeline, and displays photo evidence and the decision.
- 5. Open the four-field pre-filled Form. A real person signs in with a Google account and clicks Submit; ClaimGuard cannot perform that step.

Functional requirements

The IDs below provide a concise demo and review checklist. “Implementation anchor” points to the checked-in code; the external Form setting is identified separately.

ID	Requirement and observable behavior	Implementation anchor
R1	The team's mapping assigns exactly two distinct reference JPGs to each of 18 claims and uses all 36 files once. The validator rejects missing, duplicate, or unknown assignments; no model guesses a mapping.	data/mapping.json; lib/dataset.ts
R2	The single-column existing-claim combobox filters by claim ID or claimant name, supports keyboard or click selection, and disables unmapped entries. A selection shows incident facts, narrative, customer email, and two thumbnail previews.	components/claim-picker.tsx; components/claim-guard-app.tsx; app/api/claims/route.ts
R3	A new claim requires claimant, incident date, description, and a positive USD amount no greater than \$250,000. A blank claim ID is generated as MI-DEMO-####; location, category, and customer email are optional. TXT, MD, and DOCX description text can be appended.	components/claim-fields.tsx; lib/claim-validation.ts; app/api/claims/route.ts; lib/client/files.ts

ID	Requirement and observable behavior	Implementation anchor
R4	A new claim uses two to six JPEG, PNG, or WebP photos. Client-side ZIP extraction ignores directories/junk/non-images, permits at most 50 entries and 30 MB expanded, and rejects nested archives. Users remove extras themselves. Each lookup sends one image, with a 4,000,000-byte server cap.	components/photo-uploader.tsx; lib/client/files.ts; lib/api/upload.ts
R5	Web Detection receives base64 image bytes through the official Google client and distinguishes full, partial, and visually similar results. SHA-256 caching can avoid a repeat paid lookup. A genuine successful zero-match Vision response has status ok with zero matches, never unavailable. Conversely, a failed Vision check returned as lookup evidence has status unavailable, never a fabricated clean zero; an HTTP/database failure stops the decision and shows an error instead. A Live badge or cached timestamp identifies the result source.	lib/vision.ts; lib/lookup.ts; app/api/lookup/route.ts; components/evidence-card.tsx
R6	Cross-claim reuse is exact when SHA-256 hashes match and near when 64-bit dHash distance is at most the current cutoff of 6. A reported match shows the other claim, claimant when available, date, filename, type, and distance.	lib/hash.ts; lib/lookup.ts; lib/guardrail.config.ts; components/evidence-card.tsx
R7	The server-not browser-supplied evidence-rebuilds a ClaimEvidence value and calls the deterministic guardrail. Fewer than two photos yield Blocked with no Form link; cross-claim reuse, qualifying non-stock full web matches, or an unavailable web check can escalate. Narrative similarity alone adds context, not escalation.	app/api/decide/route.ts; lib/guardrail.ts
R8	The result area shows an agent-activity timeline, a card per photo, web and cross-claim evidence, a decision banner, reasons/notes, optional adjuster-notification status, a live recheck action, and the human-only Form handoff.	components/results.tsx; components/evidence-card.tsx; app/api/decide/route.ts
R9	The current Google Form has Collect email addresses -> Verified selected; its Settings page states that respondents must sign in to Google. The app's URL pre-fills exactly four entries-Claim ID, Claimant Name, a single-line photo-only lookup summary of at most 300 characters, and Auto-approve or Escalate. It never pre-fills an email or submits the Form.	Live Form Settings checked 2026-09-25; lib/google-form.ts; app/api/decide/route.ts
R10	MongoDB stores claims, photo records/thumbnails, Vision cache, decisions, audit entries, and daily usage. User-origin records have 14-day expiry; uploaded originals are not stored. Optional LLM identity suggestions and adjuster wording do not determine the guardrail outcome; the app works with no LLM configured.	lib/models.ts; lib/mongodb.ts; lib/llm/client.ts; app/api/decide/route.ts
R11	A daisyUI notification dock sits at the bottom right in its own layout row , so it does not cover result buttons. At most three toasts are visible; extras queue. Each auto-dismisses after about 4.5 seconds or can be dismissed manually. Toasts mark a completed decision, an unavailable web check after a decision, a recovered retry, the print dialog opening, and Start Over. An HTTP/storage lookup failure instead shows an inline error; the app does not claim that a PDF file was saved.	components/toasts.tsx; lib/client/use-toasts.ts; lib/client/use-claim-guard.ts; app/globals.css
R12	"How ClaimGuard decides" is a daisyUI collapsible section directly below the header and above the claim form, closed by default. It explains exact/near duplicates, the web-match threshold and stock-source exclusion, unavailable-check escalation, contextual narrative notes, and human Form submission. Its displayed numbers and domains come from lib/guardrail.config.ts, not duplicated copy.	components/claim-guard-app.tsx; components/how-it-works.tsx; lib/guardrail.config.ts

R5 is grounded in an observed parser bug: IMG-026's successful Vision annotation included a non-HTTP `x-raw-image://` identifier in `visuallySimilarImages`. The old parser rejected the entire annotation as invalid despite real match counts. The current parser counts that opaque item, omits it from clickable URLs/domains, accepts genuinely empty successful annotations, and keeps actual failures unavailable (lib/vision.ts; lib/vision.test.ts; docs/PHASE5.5.md).

Decision rules and stock-domain handling

Current thresholds are versioned **placeholders** for team review; they are not fraud probabilities or ground-truth labels.

Situation	Current guardrail behavior
Fewer than two photos	Blocked (R0); no completed pre-fill link.
Every claim photo matches a photo from another claim	Escalate (E1); exact or near matches are explained.
Some, but not all, claim photos match another claim	Escalate under the current PARTIAL_MATCH_ACTION setting (E2).
A photo has at least three full-image matches not classified as allow-listed stock sources	Escalate (E3). Partial and visually similar matches do not increase this count.
A web check is unavailable with no usable successful evidence	Escalate under the current fail-closed setting (E4).
Full matches exist only on allow-listed stock hosts	Add a note (N1), not a web-match escalation by itself.
Similar narrative without a photo link to that other claim; same claimant within 30 days; same amount plus similar narrative ; or the claim's first two photos resemble each other	Add context notes only (N2, N3, N4 , N5 respectively). These notes alone do not escalate.

The current stock allowlist is `pexels.com`, `pixabay.com`, and `unsplash.com`. The implementation checks the **host of each full-matching image URL**, not merely the host of a page mentioning it. It matches an exact allowed domain or a true subdomain at a DNS-label boundary: `images.pexels.com` matches `pexels.com`, but `pexels.com.evil.example` does not. A non-allow-listed website is **not blocked or accused**; its full-image matches simply count toward the threshold. Stock matches also do not guarantee approval.

Interaction notes

- Evidence cards use three distinct states: checked with matches, checked with zero matches, and unavailable/error. Only the unavailable state warns that human review is required; it never displays the clean-result reassurance.
- Choosing **Re-run live lookup** immediately clears the current decision and pre-fill link. Existing photo evidence may remain visible with a “current decision has not been completed” message while the new lookup runs; once all required results are available, the server decides again.
- **Open pre-filled form** opens a new tab for the human. **Copy summary** copies the decision text. **Save as PDF** invokes the browser print dialog; ClaimGuard cannot tell whether the person actually saved a PDF. **Start Over** resets on-screen state only.
- The decision timeline marks the final human sign-in/Submit step **pending** when a pre-fill link is generated. Optional AI wording is supplementary and cannot change the deterministic decision.
- The bottom-right toast dock reports meaningful outcomes/recovery without overlaying actions. A failed HTTP/storage lookup appears as an inline error; an unavailable web result can produce a warning toast after the guardrail finishes.

Operational and privacy requirements

ID	Requirement and implementation basis
NFR1	Each API route uses the Node runtime with a 10-second maxDuration. The browser extracts ZIPs and sends one image per lookup request; it never posts a complete ZIP or photo batch. (app/api/*/route.ts, lib/client/files.ts, lib/client/use-claim-guard.ts)
NFR2	Secrets stay in server-side environment variables. Reference originals are outside public/; the photo endpoint serves stored JPEG thumbnails only. The page sets noindex metadata and robots.txt disallows crawling. (lib/env.ts, app/api/photos/[id]/thumbnail/route.ts, app/layout.tsx, app/robots.ts)
NFR3	Safe transient MongoDB operations get bounded retries; the browser automatically retries one retryable database 503 for claims, lookup, or decision, then shows an inline error with Try again. Failed Vision checks remain unavailable evidence. Daily service limits and a best-effort per-IP lookup limiter bound use. (lib/mongo-retry.ts, lib/client/api.ts, components/claim-guard-app.tsx, lib/api/security.ts)
NFR4	The claim picker uses keyboard-aware combobox behavior; controls expose labels and status text. Layouts stack on narrow screens. Reduced-motion preferences disable toast/collapse animation. (components/claim-picker.tsx, components/claim-guard-app.tsx, app/globals.css)
NFR5	User-origin claims, photos, decisions, and audit records carry 14-day expiry; uploaded originals are not persisted. Start Over clears only client-side form/results state and does not delete database records. (lib/models.ts, lib/mongodb.ts, lib/client/use-claim-guard.ts)

Demonstration acceptance checks

For the course demo, the team should show a valid 18-to-36 mapping, a real one-photo Web Detection lookup, an existing-claim decision with its timeline and evidence, the clean-zero versus unavailable distinction, and the human-only pre-filled Form. If showing an unavailable web check, it should lead to E4 escalation, not a reassuring clean card. A zero-LLM configuration must remain functional. The team should present its threshold rationale separately; the current numeric values are placeholders.

Implemented interface screenshots

Figures 1-5 are actual captures of the deployed ClaimGuard app and the signed-in Google Form on September 25-28, 2026, not mockups. Figures 1 and 3 show MI-10237 (Devon Price); Figure 4 shows MI-10236 (Ana Torres). Those production app results used cached Vision evidence. Figure 6 is a deliberately simulated failure in isolated local development, not production. No Google Form was submitted. Screenshots are cropped to the relevant sections; the Form account address is redacted for privacy.

Figure 1 - Existing claim selection and mapped photos

Production app showing MI-10237 selected, its claim details, and two mapped photo thumbnails

Reference claim

MI-10237 · Devon Price · \$2,450

MI-10237 · Devon Price

Incident date

2026-08-07

Amount

\$2,450.00

Location

Garland, TX

Category

Comprehensive

Customer email

devon.price@email.com

Claim description

Hail damage to hood and roof during the Aug 6 storm system.

2. Mapped claim photos



IMG-031.jpg

queued



IMG-026.jpg

queued

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5

Figure 2a - New claim fields

Production app showing new-claim fields

Tell us about the claim

Claim ID (optional)

Claimant name

Blank IDs become MI-DEMO-####.

Incident date

mm/dd/yyyy

Claim amount (USD)

Greater than \$0, up to \$250,000.

Location (optional)

Category (optional)

Customer email (optional)

Claim context only. Never used to fill the Google account email field.

Claim description

0 / 10,000 characters

Or import description text

Choose File

No file chosen

TXT, Markdown, or Word · up to 3 MB. Imported text is appended, not substituted.

Figure 2b - Photo upload

Production app showing the photo upload area

2. Add claim photos

Choose 2–6 photos. ZIPs are opened on this device; only one image is sent per request.

Drop images or a ZIP here

JPG, PNG, WebP · max 50 archive entries / 30 MB expanded · no nested archives

Choose Files

No file chosen

Files up to 3 MB keep their original bytes. Larger files are resized to a maximum 1600px edge before hashing and upload.

Figure 3a - Agent activity timeline

Production app showing the agent activity timeline, ending with human submission pending

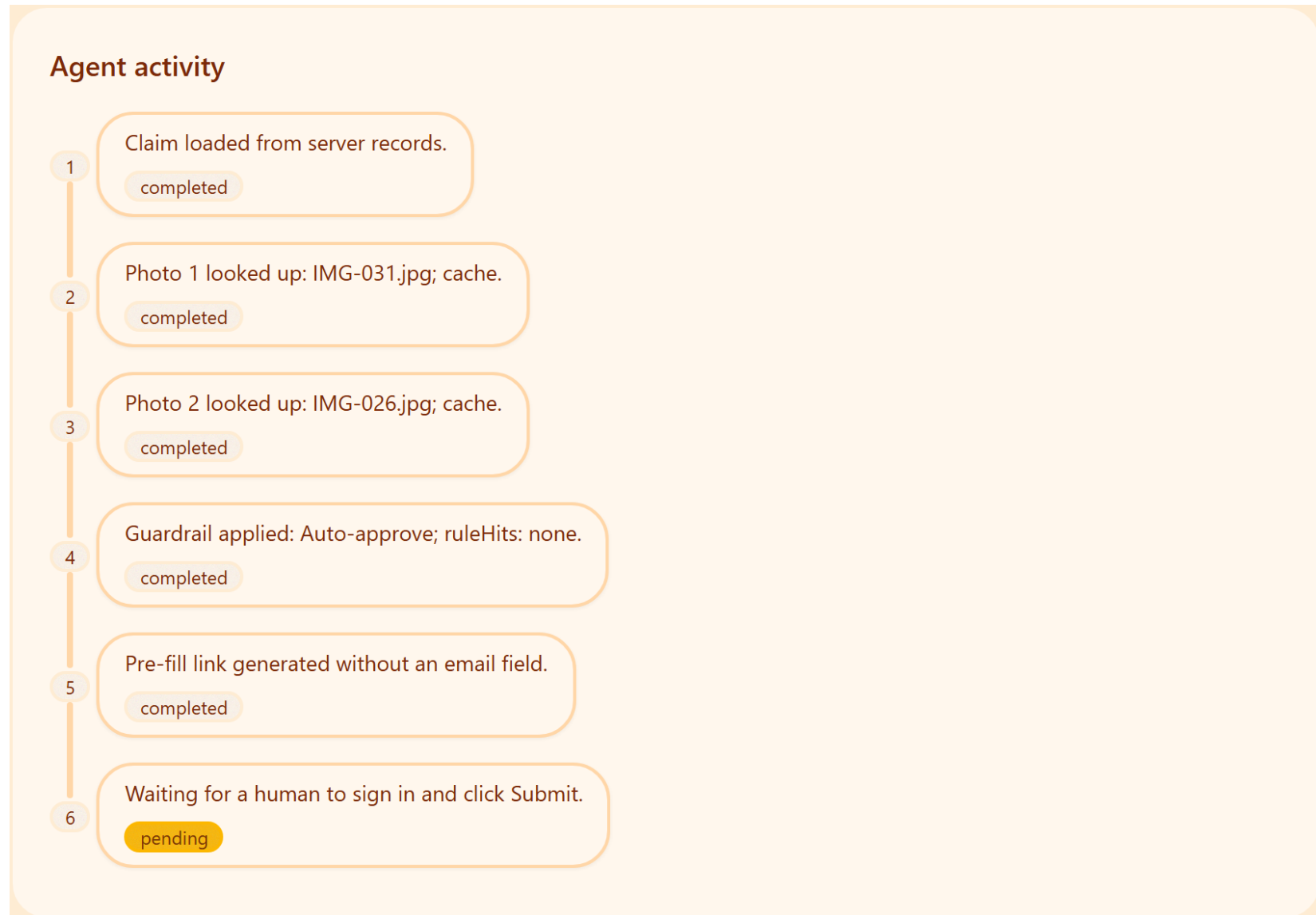


Figure 3b - Photo evidence

Production app showing the two Web Detection evidence cards



IMG-031.jpg

Cached at 9/23/2026, 9:26:44 PM

SHA-256 cdbd4ae8b83bf26a...

Full matches	Non-stock full
1	1
Partial matches	Similar images
0	20

Top reported domains

lookaside.fbsbx.com Not allow-listed

youtube.com Not allow-listed

facebook.com Not allow-listed

pbs.twimg.com Not allow-listed

cdn.auto-swiat.pl Not allow-listed

Cross-claim matches

No matches to other claims.

Web checked: matches found

A similar image is not proof of reuse. An allow-listed domain is context, not an approval.

Re-run live lookup



IMG-026.jpg

Cached at 9/23/2026, 10:04:57 PM

SHA-256 5bb9edd32f4f9647...

Full matches	Non-stock full
0	0
Partial matches	Similar images
1	20

Top reported domains

youtube.com Not allow-listed

us.amazon.com Not allow-listed

marchomecenter.com Not allow-listed

amazon.com Not allow-listed

enrichfin.com Not allow-listed

Cross-claim matches

No matches to other claims.

Web checked: matches found

A similar image is not proof of reuse. An allow-listed domain is context, not an approval.

Re-run live lookup

Figure 3c - Decision and Form handoff

Production app showing the Auto-approve banner and human-only form actions

Guardrail decision

Auto-approve

Rule version 1.0.0 · human submission still required

Reasons

No escalation rule was triggered. This does not prove the photos are original.

Photo lookup summary

Photo 1 (IMG-031.jpg): full 1, non-stock 1, partial 0, similar 20; other-claim matches 0. | Photo 2 (IMG-026.jpg): full 0, non-stock 0, partial 1, similar 20; other-claim matches 0.

You must sign in with your own Google account and click Submit. The agent never submits for you.

Open pre-filled form 

Copy summary

Save as PDF

Start Over

This clears the form on your screen. It doesn't delete anything from the database.

Adjuster notification: Not needed for this decision.

Figure 4 - Escalate example: MI-10236 (Ana Torres)

The production app escalated this \$4,100 claim because both mapped photos exactly match photos assigned to MI-10250. The same amount and similar narrative appear only as a note for review, not as the escalation trigger.

Production app showing Ana Torres's Escalate decision, exact cross-claim photo matches, and human-only Form handoff

Guardrail decision

Escalate

Rule version 1.0.0 · human submission still required

Reasons

- IMG-033.jpg (exact match, distance 0) matches IMG-004.jpg from claim MI-10250 (2026-08-20).
- IMG-020.jpg (exact match, distance 0) matches IMG-024.jpg from claim MI-10250 (2026-08-20).

Notes for review

- Claim MI-10250 has the same amount (\$4100) and a similar narrative.

Photo lookup summary

Photo 1 (IMG-033.jpg): full 0, non-stock 0, partial 0, similar 20; other-claim matches 1. | Photo 2 (IMG-020.jpg): full 0, non-stock 0, partial 0, similar 20; other-claim matches 1.

You must sign in with your own Google account and click Submit. The agent never submits for you.

Open pre-filled form 

Copy summary

Save as PDF

Start Over

This clears the form on your screen. It doesn't delete anything from the database.

Adjuster notification: No adjuster-notification webhook is configured. Share the findings with an adjuster manually.

Figure 5 - Pre-filled Google Form in Edge

This is the real signed-in Form reached from Ana's result. The claim ID, claimant, and photo lookup summary are pre-filled; the Form also selected Escalate below the captured viewport. The email address shown by Google is redacted in this image, and the email confirmation was left unchecked. A human must complete that step and click Submit.

Signed-in Google Form in Edge with Ana Torres claim fields pre-filled and account email redacted

The screenshot shows a Google Form titled "ClaimGuard Adjuster Intake". At the top, there is a header bar with a blue background. Below the header, the form is divided into several sections. The first section contains a redacted email address followed by a link that says "witch account" and a small icon of a person. Below this is a red asterisk followed by the text "* Indicates required question". The second section is titled "Email *" and contains a checkbox labeled "Record [redacted] as the email to be included with my response". The third section is titled "Claim ID *" and contains the text "MI-10236". The fourth section is titled "Claimant Name *" and contains the text "Ana Torres". The fifth section is titled "What the lookup found *" and contains two lines of text: "Photo 1 (IMG-033.jpg): full 0, non-stock 0, partial 0, similar 20; other-claim matches 1. |" and "Photo 2 (IMG-020.jpg): full 0, non-stock 0, partial 0, similar 20; other-claim matches 1. |".

ClaimGuard Adjuster Intake

[redacted] witch account

* Indicates required question

Email *

☐ Record [redacted] as the email to be included with my response

Claim ID *

MI-10236

Claimant Name *

Ana Torres

What the lookup found *

Photo 1 (IMG-033.jpg): full 0, non-stock 0, partial 0, similar 20; other-claim matches 1. |
Photo 2 (IMG-020.jpg): full 0, non-stock 0, partial 0, similar 20; other-claim matches 1. |

Figure 6 - Deliberately simulated unavailable web check and E4 escalation

On September 28, an isolated local run used an invalid Vision API key. Google returned HTTP 400 with “API key not valid,” so both photo checks for MI-10237 were genuinely unavailable and the guardrail escalated. The evidence-card alert and decision below are two cropped views from that same run, not a naturally occurring dataset failure or a production issue. Re-running MI-10237 in production with its previously cached successful Vision evidence returned Auto-approve (Figure 3c); the comparison did not make a new paid Vision call.

Local simulated E4 evidence card showing Web check unavailable, HTTP 400, and the human-review warning

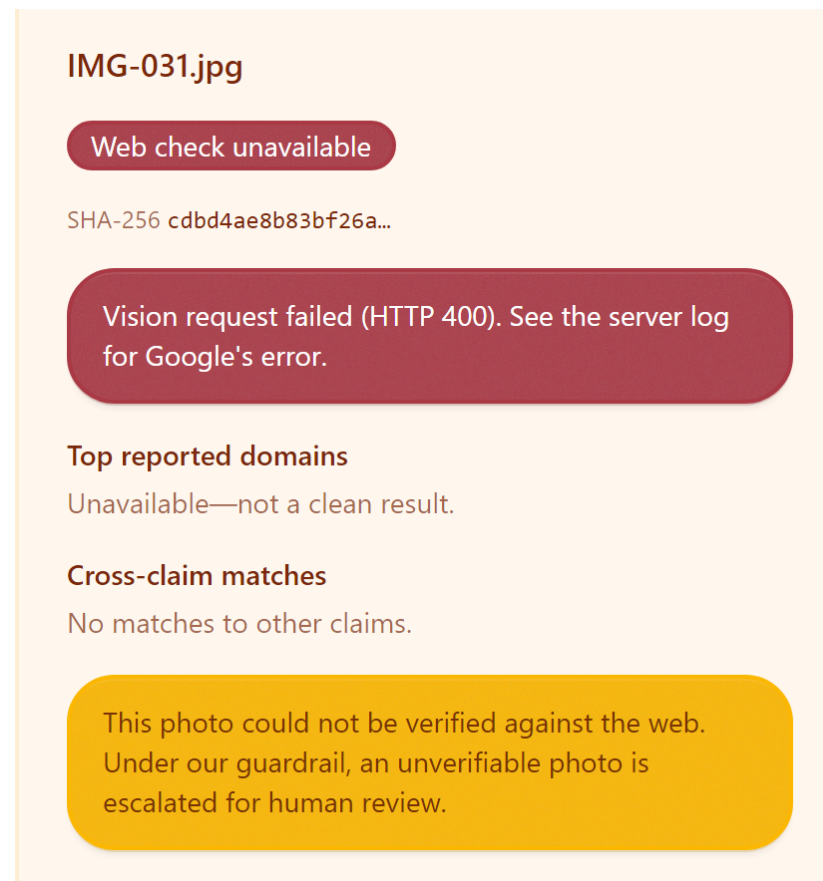


Figure 6 (continued) - E4 decision for the same local run

Same local simulated E4 run showing the Escalate banner and unavailable-web-check reasons

Guardrail decision

Escalate

Rule version 1.0.0 · human submission still required

Reasons

- IMG-031.jpg could not be checked against the web (service unavailable); escalating to be safe.
- IMG-026.jpg could not be checked against the web (service unavailable); escalating to be safe.